Mr Roberto Gradara

robertos.gradara@gmail.com • +39-328-972-0280 • LinkedIn • Visa: *Italian Passport*

Summary

Current PhD student in Mathematics (Probability/Statistics) with industrial research experience in mathematical finance. Attended the "Toppo Wasserman" School of Advanced Studies. Completed projects in ML (e.g., Geometric Deep Learning, Gaussian Processes) and numerical simulations (e.g., PDEs, stochastic processes). Fluent in Python and other languages. Served as a teaching assistant for several undergraduate modules.

Education

Imperial College London — London, UK

PhD, Mathematics — *09/2025 – now*

PhD research in Mathematics (Probability/Statistics), focusing on random measures and stochastic PDEs, under the supervision of Prof. R. Passeggeri. Position fully funded by the Roth Scholarship with 3.5-year stipend. Currently serving as Graduate Teaching Assistant / Teaching Assistant for the undergraduate course *Calculus and Applications*.

University of Udine — Udine, IT

MSc, Mathematics - 110 *cum laude*/110 (Distinction) — *12/2021 – 03/2024*

Thesis Title: *Bounds for exotic options*.

Overview: Applied Villani's approach to optimal transport in measure spaces and implemented a novel setup for path-dependent option pricing. Developed an efficient object-oriented Python code.

Supervisors: Prof A. Zanette, Dr C. Martini, Prof A. Molent.

Modules: Completed units in the fields of Geometry (29/30, 30 *cum laude*/30, 28/30), Nonlinear Analysis & Modeling (30 *cum laude*/30), Computational Methods (29/30), Statistics (27/30, 30/30), and Times Series Analysis (30 *cum laude*/30).

"Toppo Wasserman" School of Advanced Studies — Udine, IT

MSc, Mathematics Excellence Program - 110/110 (Distinction) — *09/2018 – 03/2024*

Description: Higher education institution modeled after the French grandes écoles. It complements the University of Udine curriculum by offering supplementary courses aimed at training students to academic research. This program awards an additional diploma equivalent to a MSc. Admission is subject to a competitive exam and requires maintaining excellent academic performance through the years. A full scholarship (covering living expenses, accommodation, and University fees) is included.

Activities: See "Computational Projects" section for details.

University of Udine — Udine, IT

BSc, Mathematics - 110 *cum laude*/110 (First Class Honours) — *09/2018 – 12/2021*

Dissertation Title: *Conservative Systems generated by Delay Differential Equations.*

Overview: Applied the Kaplan-Yorke approach to transform specific DDEs into equivalent Hamiltonian systems for numerical computation and studied their asymptotic properties. Contributed by generalizing the main theorem presented by Kaplan-Yorke, weakening one of its key assumptions.

Supervisor: Prof. F. Zanolin.

Liceo A. Righi — Cesena, IT

Scientific High School Diploma (Cambridge Curriculum) - 98/100 — *2013 – 2018*

Description: Completed a "Liceo Scientifico" diploma. Cambridge Curriculum is a special path comprising additional weekly lessons in English. Achieved IGCSE exam results: Mathematics (A*), Physics (A*), Biology (A).

Awards

- **Mathematical Team-Olympiad Competition:** Qualified (2017, 2018) for the National final.
- **National mental math championship:** Achieved top 50 in the Italian championship.

Publications

- Hedging options with other options in a trinomial tree: theory.
  R. Gradara, C. Martini
  In Preparation

Computational Projects

Pricing Basket Options — Udine, IT
Student at University of Udine — *10/2023*
Overview: Exposition of the theory for simulating correlated Gaussian processes and tree-based methods. Implemented relevant analytical methods and Monte Carlo simulations using Python and C.
Download: Google Drive file (click here).

Mathematical Foundations of Deep Learning — Udine, IT
Individual Project for "Toppo Wasserman" School of Advanced Studies — *03/2023*
Overview: Research project focused on the mathematical underpinnings of machine learning and deep learning, with an emphasis on risk curves and random matrices.

- Developed a formal framework for Machine Learning by proving theorems supporting the classical U-shape risk curve and the double descent curve in modern Deep Learning.
- Provided a mathematical motivation for using random matrices, introduced key concepts like the Marcenko-Pastur law, and discussed dimensionality challenges.
- Step-by-step description of the relevant Python implementation to practically demonstrate the learning models' capabilities.

Download: Google Drive file (click here).

Empirical Anlysis: *PLUG Power* time series — Udine, IT
Student at University of Udine — *07/2022*
Overview: Search for both the best fit and the forecast of time series using: EDA, in-depth discussion, and evaluation of linear (ARIMA + GARCH) and nonlinear (SETAR, Neural Network) models.
Download: Google Drive file (click here).

Numerical Approximation of Partial Differential Equations — Udine, IT
Student at University of Udine — *07/2022*
Overview: Characterisation and implementation of the finite element Galerkin method in MATLAB for 1D and 2D PDEs. Ensured the numerical stability and precision of the solutions.
Download: Google Drive file (click here).

SIR Model with Delay — Cetraro, IT
Attendeed at "Mathematical Modelling for Epidemiology" Summer School — *09/2022*
Overview: Selected to join a research project on population dynamics alongside two professors and two senior students. Applied optimal transport theory and semigroup theory to delay differential equations in epidemiological models. This experience deepened my expertise in mathematical modeling.

Conferences

Optimal Transport and Model Calibration Workshop — Marseille, FR
Conference Lecturer — *04/2024*
Overview: Scientific retreat featuring leading experts such as J-D. Benamou, F-X. Vialard, M. Hoffman, S. Crepey, and G. Loeper.

- Co-presented a lecture with my supervisor (Dr C. Martini) on Martingale Optimal Transport, detailing the state-of-the-art results of our ongoing research on path-dependent option pricing.

First Italian Summer School on Geometric Deep Learning — Pescara, IT
Summer school attendance, Speaker — *07/2022*
Overview: Scientific retreat featuring leading experts such as Michael Bronstein, C. Bodnar, F. Di Giovanni, P. De Haan, and M. Weiler.

- The experience strongly stimulated further curiosity about GDL, leading to co-presenting related topics in a supplementary university course lecture a few months later.

Work Experience

Reply Data — Fully Remote, IT

Data science — *09/2024 – Present*

Description: Consulting firm providing analytics and AI-driven data services across various industries.

- Leading the development of an AI chatbot proof of concept for a luxury fashion brand's website, designed to answer product-specific questions and suggest complementary pairings.
- Main tasks include conducting EDA and producing metrics to identify the optimal embedding method for maximizing the model's performance in finding the relevant documents (among the available ones) to user's queries, prompt engineering to improve further the documents retrieval and the answer's generation (RAG).

Keywords: `Machine Learning Engineering, Large Language Models (LLM), API Integration`

Zeliade Systems — Paris, FR

Researcher — *10/2023 – 07/2024*

Description: Provider of technology and consultancy services to Central Counterparty Clearing Houses worldwide, in valuation, risk management, and regulatory compliance.

- Replicated the results of a recent paper which proposes a Machine Learning algorithm to generate synthetic time series data by leveraging on optimal transport theory in measure space.
- Contributed to the validation team with data-related tasks in Pandas.
- Completed the MSc thesis project in Mathematical Finance "*Bounds for exotic options*" and lectured at the "*Optimal Transport and Model Calibration*" Workshop.
- Preliminary results are available in the MSc Thesis and are currently being organised for publication.

Keywords: `Stochastic Processes, Quantitative Finance, Analytical Thinking, Algorithm Development`

University of Udine — Udine, IT

Teaching Assistant — *02/2023 – 09/2023*

- Conducted exercises for Geometry 1 and Analysis 1 in the Mathematics degree program, and Analysis 2 in the Civil Engineering degree program.

Keywords: `In-Depth Subject Knowledge, Pedagogical and Communication Skills, Assessment and Evaluation, Time Management`

Mathesis — Udine, IT

Speaker and Teacher — *03/2020, 03/2023, 03/2024*

Description: National association dedicated to promoting mathematics education through high school competition and to preparing students for the advanced stages of the Mathematics Olympiads.

- Delivered talks teaching problem-solving techniques and demonstrating how to apply foundational theoretical concepts to complex problems.
- Proposed two problems for the National Training stage of the Mathematics Team-Olympiad Competition.

Keywords: `Math Competitions, Problem-Solving Techniques, Educational Workshops`

Languages

▷ **English:** Bilingual Proficiency (level C1 certificate by Cambridge CAE, 2017)

▷ **Italian:** Native

▷ **French:** Basic